Tingsong Xiao

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Homepage; Google Scholar

Research Interests

AI for Health, Data Mining, Time Series, and Event Sequence Modeling, LLMs for Medical AI, Medical Image, and Disease Progression Modeling.

Education

Doctor of Philosophy, Ph.D. | **GPA: 3.86/4.00** | ***Computer Science*** — Aug. 2022 – Now
University of Florida — Gainesville, FL, U.S.

Master of Science, M.S. | **GPA: 3.86/4.00** | ***Computer Science*** — Aug. 2022 – Dec. 2024
University of Florida (Conferred during Ph.D. program) — Gainesville, FL, U.S.

Bachelor of Engineering, B.E. | **GPA: 3.92/4.00** | ***Computer Science*** — Aug. 2017 – June 2021
University of Electronic Science and Technology of China (UESTC) — Chengdu, China

Experience

Applied Researcher Intern — May 2025 – Aug. 2025
eBay Inc. — San Jose, CA, U.S.

- Developed a **multi-modal LLM** pipeline to extract structured knowledge from **unstructured text and image-based product data**, addressing complex information extraction at scale.
- Built a **post-training** framework using **Parameter Efficient Fine-tuning** (LoRA) and multi-strategy distillation (response-based and logit-based) from state-of-the-art **VLMs and LLMs** into compact student models.
- Developed an autonomous **LLM agentic workflow** to synthesize high-fidelity training datasets by mapping raw descriptions to structured aspect-value pairs, significantly reducing reliance on manual labeling and improving model robustness on rare data distributions.
- Delivered a compact model reaching $>$ **90% F1** across 10 business-critical categories while reducing inference latency by $>$ **5**$\times$, enabling deployment **at scale** to millions of listings.

Research on Continuous-Time Disease Progression Modeling — May 2024 – May 2025
University of Florida — Gainesville, FL, U.S.

- Addressed **continuous-time disease progression modeling** from longitudinal EHRs to enable early risk detection and patient sub-phenotyping under **irregular visit times** and **patient heterogeneity** (e.g., type 2 diabetes and cardiovascular complications).
- Proposed **TD-HNODE**, integrating **hypergraph representation learning** from clinical trajectories with a **Neural ODE** to learn continuous-time dynamics of comorbidities, biomarkers, and medical events.
- Outperformed state-of-the-art baselines on two large-scale real-world clinical datasets in predicting health deterioration patterns. This work was published at top AI conference **ICLR 2026.**

Research on Irregular Clinical Event Sequence Modeling — Aug. 2023 – May 2024
University of Florida — Gainesville, FL, U.S.

- Addressed **irregular health time-series and event sequence** prediction in real-world digital health settings (wearable sensors / mobile health), where signals are long-horizon, sparse, and non-uniformly sampled.
- Developed **XTSFormer**, an **efficient Transformer-based** Temporal Point Process model for irregular timed clinical events, which captured **multi-scale event interactions** while remaining **computationally efficient** for long event sequence.
- Achieved **5**$\times$ **faster** training on long sequences (up to 100k length) and improved predictive accuracy/F1 by **3.3%** on large-scale longitudinal health datasets. This work was published at top AI conference **AAAI 2025.**

Research on Causal Treatment Effect Estimation in Digital Health — Aug. 2025 – Present
University of Florida — Gainesville, FL, U.S.

- Addressed the critical need for real-world comparative effectiveness research in Heart Failure (HFpEF) management where standard randomized controlled trials (RCTs) for multi-drug regimens are infeasible , focusing on estimating treatment effects from large-scale, heterogeneous EHR data.

- Developed HyperDR, a **hypergraph-based causal inference** framework that represents drug classes as nodes and multi-drug regimens as hyperedges to capture high-order **drug-drug interactions**; integrated a **Doubly Robust** (DR) objective with an augmented inverse-probability-weighted regularizer to ensure unbiased estimation even under selection bias or model failure.
- Improved hospitalization **risk prediction** and achieved superior covariate balance across diverse patient subgroups (segmented by sex, race, and obesity) , providing clinically validated **treatment rankings** to support personalized digital therapeutics and risk stratification.

SELECTED PUBLICATIONS

- **Tingsong Xiao**, Yao An Lee, et al. ”Temporally Detailed Hypergraph Neural ODE for Type 2 Diabetes Progression Modeling.” *The Fourteenth International Conference on Learning Representations (ICLR 2026).*
- **Tingsong Xiao**, Zelin Xu, et al. ”XTSFormer: Cross-Temporal-Scale Transformer for Irregular-Time Event Prediction in Clinical Applications.” *The 39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025).*
- **Tingsong Xiao**, Yu Huang, et al. ”Large Language Models for Longitudinal Electronic Health Records: A Survey.” *medRxiv 2026.1.29.345087*
- **Tingsong Xiao**, Yao An Lee, et al. ”Hypergraph-Based Doubly Robust Estimation for Causal Inference of Drug Combination Effects in Heart Failure Treatment.” *medRxiv 2025.12.11.25342026*
- **Tingsong Xiao**, Lu Zeng, Xiaoshuang Shi, Xiaofeng Zhu, and Guorong Wu. ”Dual-graph Learning Convolutional Networks for Interpretable Alzheimer’s Disease Diagnosis.” *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2022). Springer, Cham, 2022.* **Early Accepted Oral Paper.**
- Zelin Xu, Yupu Zhang, **Tingsong Xiao**, et al. ”CoastalBench: A Decade-Long High-Resolution Dataset to Emulate Complex Coastal Processes.” *The Forty-Second International Conference on Machine Learning (ICML 2025).*
- Yupu Zhang, Zelin Xu, **Tingsong Xiao**, et al. ”DecoyDB: A Dataset for Graph Contrastive Learning in Protein-Ligand Binding Affinity Prediction.” *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025).*
- Wenchong He, Zhe Jiang, and **Tingsong Xiao**, et al. ”A survey on uncertainty quantification methods for deep learning.” *ACM Computing Surveys (2025)*.
- Zelin Xu, **Tingsong Xiao**, Wenchong He, et al. ”Spatial-Logic-Aware Weakly Supervised Learning for Flood Mapping on Earth Imagery.” *The 38th Annual AAAI Conference on Artificial Intelligence (AAAI 2024).*
- Wenchong He, Zhe Jiang, and **Tingsong Xiao**, et al. ”A Hierarchical Spatial Transformer for Large Numbers of Point Samples in Continuous Space.” *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*.
- Zelin Xu, **Tingsong Xiao**, Wenchong He, Yu Wang, and Zhe Jiang. ”Spatial Knowledge-Infused Hierarchical Learning: An Application in Flood Mapping on Earth Imagery.” *ACM International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL 2023).* **Best Paper Award.**
- Jing Hu, Xincheng Wang and Ziheng Liao, **Tingsong Xiao***. ”L-GCN: Local Graph Neural Network for 3D Point Clouds Classification.” *IEEE International Conference on Multimedia and Expo (ICME 2023).* **(* corresponding author)**.
- Lu Zeng, Hengxin Li, **Tingsong Xiao**, Fumin Shen and Zhi Zhong. ”Graph convolutional network with sample and feature weights for Alzheimer’s disease diagnosis.” *Information Processing & Management (IP&M) 59.4 (2022): 102952.*
- Xiaochen Wang, **Tingsong Xiao**, Jie Tan, Deqiang Ouyang and Jie Shao. ”MRMRP: multi-source review-based model for rating prediction.” *International Conference on Database Systems for Advanced Applications (DASFAA). Springer, Cham, 2020.*

- Xiaochen Wang, **Tingsong Xiao** and Jie Shao. "EMRM: Enhanced Multi-source Review-Based Model for Rating Prediction." *International Conference on Knowledge Science, Engineering and Management (KSEM). Springer, Cham, 2021.*

HONORS AND AWARDS

AAAI Scholarship and Volunteer Award Feb. 2025
To recognize outstanding contributions and active participation through academic excellence in the 39th AAAI Conference on Artificial Intelligence (AAAI-25).

SDM Doctoral Forum Award Mar. 2024
A competitive NSF-funded venue recognizing promising data mining researchers for their intellectual merit and broader impacts.

ACM SIGSPATIAL Best Paper Award Nov. 2023
Awarded annually to the best paper at the ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems.

Gartner Group Graduate Fellowship Apr. 2023, Apr. 2025
Recognizes outstanding graduate students at the University of Florida for academic excellence and research potential.

MICCAI 2022 Student Travel Award Mar. 2022
To reward first author students of the highest-quality MICCAI papers.

INVITED JOURNAL AND CONFERENCE REVIEWERS

- Information Sciences
- Neural Networks
- Neural Processing Letters
- Pattern Recognition Letters
- IEEE International Symposium on Biomedical Imaging (ISBI)
- IEEE International Conference on Data Mining (ICDM)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Conference on Neural Information Processing Systems (NeurIPS)
- Association for the Advancement of Artificial Intelligence (AAAI)
- International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)
- ACM Transactions on Intelligent Systems and Technology (TIST)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Computational Social Systems (TCSS)
- IEEE Transactions on Multimedia (TMM)
- Information Processing & Management (IP&M)
- Future Generation Computer Systems